

Uniform dependence on the number of controlled prefix moments

Complete proof; three independent mathematical reviews completed

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This note makes the fixed-parameter constants in `../rational_designs/asymptotic/multiple_prefix_moment_conditioning.tex` explicit enough for a growing-number-of-dice audit. It does not alone supply a uniform complexity bound for the entire construction. All norms are Lebesgue $L^2[0, 1]$ norms.

Lemma 1 (A uniform multiplier bound). *Let Q be a real polynomial of degree h , with all roots in $[0, 1]$ and leading coefficient of absolute value at least one. Then for $p \in \mathcal{P}_r$,*

$$\|Qp\|_2 \geq 2^{-1/2} [4h(r+1)^2]^{-h} \|p\|_2$$

when $h \geq 1$.

Proof. Remove intervals of radius $\varepsilon = [4h(r+1)^2]^{-1}$ around the roots, counted with multiplicity. Their total length is at most $2h\varepsilon$. The polynomial evaluation inequality $\|p\|_\infty \leq (r+1)\|p\|_2$ shows that at most half of its squared norm is lost. On the remainder, factorization gives $|Q| \geq \varepsilon^h$. Integrating there proves the claim. No root-separation bound is needed. $\square$

Theorem 1. *There is an absolute constant C with the following property. Use $q \geq 1$ Legendre bands and cumulative moments $M_0, \dots, M_q$ from the cited multiple-prefix note, with multiplier degree r , total band degree D , and $0 < \eta \leq 1/(2q)$. Then*

$$\left(\|p_*\|_2^2 + \sum_{l=0}^q \|p_l\|_2^2 \right)^{1/2} \leq e^{C(q+1)^2 \log(q+2)} (D+1)^{C(q+1)^2} \eta^{-2q} \left\| p_* + \sum_{l=0}^q p_l M_l \right\|_2.$$

Minimal band choices satisfy $D+1 \leq C3^q(r+q+2)$. In particular, if $r, q, \eta^{-1} \leq P$ and $P \geq 2$, the inverse norm is at most $P^{C(q+1)^3}$. More generally it is at most $P^{C(q+1)^3} \eta^{-2q}$ when only $r, q \leq P$.

Proof. We track the existing proof, retaining its notation $w = t(1-t)$, $H_i = \int_0^t P_{d_i}(2u-1)du$, $\lambda_i = d_i(d_i+1)$, and $\mathcal{L}_i C = (wC)'' + \lambda_i C$. The iterated three-band estimate costs at most

$$T = C_0^q (D+1)^{6q} \eta^{-q}.$$

Indeed each iteration is the same absolute-constant three-band lemma; summing squares does not introduce another dimension-dependent factor. For $F = p_* + \sum_l p_l M_l$, it bounds every coefficient in $F = P + \eta \sum_i (H_i A_i + H'_i B_i)$ by $T\|F\|_2$. The exact primitive formulas give

$$-B_i = w \sum_{l=1}^q l p_l \mathcal{L}_i^{-1} t^{l-1}.$$

Choose the standard shifted Jacobi polynomials $J_j(t) = P_j^{(1,1)}(2t-1)$, $0 \leq j < q$, and put $\mu_j = (j+1)(j+2)$. Then $\mathcal{L}_i J_j = (\lambda_i - \mu_j) J_j$. Writing $t^{l-1} = \sum_j a_{jl} J_j$ and $c_j = \sum_{l=j+1}^q l a_{jl} p_l$ gives

$$-B_i = \sum_{j=0}^{q-1} \frac{w J_j c_j}{\lambda_i - \mu_j}.$$

All nonzero eigenvalue differences are integers, and $|\lambda_i - \mu_j| \leq 3(D+1)^2$. The Cauchy determinant formula therefore bounds the inverse matrix by

$$\|C^{-1}\| \leq q! 3^{q^2} (D+1)^{2q^2},$$

up to an additional factor polynomial in q which is harmless below: the determinant numerator is a nonzero integer, while each entry has absolute value at most one and each cofactor at most $(q-1)!$. Consequently all $\|wJ_j c_j\|_2$ are bounded by that quantity times $qT\|F\|_2$.

Every J_j has its j roots in $(0, 1)$ and leading coefficient $\binom{2j+2}{j} \geq 1$. Thus wJ_j satisfies Lemma 1 with degree $j+2 \leq q+1$. Recovering c_j costs at most

$$\sqrt{2}[4(q+2)(r+1)^2]^{q+2}.$$

The triangular transformation between $(p_1, \dots, p_q)$ and the c_j has inverse norm at most $\exp(Cq^2 \log(q+2))$. For completeness, orthogonality with weight w and

$$\int_0^1 wJ_j^2 = \frac{j+1}{(2j+3)(j+2)}$$

give $|a_{jl}| \leq C\sqrt{q+1}$. The diagonal coefficient $a_{l-1,l}$ is the reciprocal of the leading coefficient of J_{l-1} , so it is at least 4^{-l} . The determinant/cofactor formula, for entries la_{jl} , gives the asserted inverse bound directly.

We have now bounded all p_l , $l \geq 1$, by

$$e^{C(q+1)^2 \log(q+2)} (D+1)^{2q^2+8q+4} \eta^{-q} \|F\|_2,$$

after increasing C to absorb factors depending only on q . Since $0 \leq M_l \leq 1$, subtract those terms. The remainder is $p_* + p_0 M_0$; its trinary representation has polynomial coefficient $p_* + tp_0$ and coefficients $-p_0$ at each H_i . One further application of the same band estimate costs another T . This proves the stated bound with an absolute C .

For minimal bands, writing $E_i = r + q + 1 + \sum_{j \leq i} (d_j + 1)$ gives $E_i \leq 3E_{i-1} + C$, hence the displayed bound on D . Finally $\log(D+1) \leq Cq + \bar{C} \log P$ when $r, q \leq P$. All constant factors above can therefore be absorbed into $P^{C(q+1)^3}$, with an extra η^{-2q} if it is not bounded by P . $\square$

The same estimates hold for a density $a \leq w_0 \leq b$, with the explicit additional norm-comparison factors $a^{-1/2}, b^{1/2}$. They also hold on an exact rule for the relevant squared polynomials. Thus this conditioning step contains no hidden exponential dependence on the polynomial multiplier degree.